Fernanda Lang Schumacher

Assistant Professor, Division of Biostatistics
College of Public Health, The Ohio State University

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Education

2018–2021 **Ph.D. in Statistics.** State University of Campinas (UNICAMP), Campinas, SP, Brazil. Advisors: Larissa Avila Matos and Victor Hugo Lachos Davila.

2015–2016 **M.Sc. in Statistics.** State University of Campinas (UNICAMP), Campinas, SP, Brazil. Advisors: Victor Hugo Lachos Davila and Filidor Vilca Labra.

2011–2014 **B.S. in Statistics.** State University of Maringa (UEM), Maringa, PR, Brazil. Graduated with honors.

Professional experience

04/2022–present **Assistant Professor**, Division of Biostatistics, College of Public Health, The Ohio State University, Columbus, OH, USA.

11/2016–02/2018 **Statistician**, Technologist in geographic and statistical information, Brazilian Institute of Geography and Statistics (IBGE), Rio de Janeiro, RJ, Brazil.

08/2017–12/2017 **Instructor**, Department of Statistics, National School of Statistical Sciences (ENCE), Rio de Janeiro, RJ, Brazil.

08/2013–07/2014 **Project director and founding member**, *Estats Consultoria* (junior company of statistical consulting), UEM, Maringa, PR, Brazil.

Prizes and awards

2026 IBS-CWS Florence Nightingale Award.

2025 Jacob Palis Award (Fulbright Brazil). In collaboration with Rojas Guerra, R., Figard, R., Fontana Ribeiro, T., and Pena-Ramirez, F. A.

2024 CAPES visiting scholar travel award.

2023 IMS New Researchers Conference travel award.

2021 Winner of the competition for students of statistics in association with the Conference in Honour of Fred Smith and Chris Skinner.

2018 Second place, Best Master's Thesis contest, 23rd SINAPE.

Research interests

- Censored regression models
- DNA methylation patterns
- Linear and nonlinear mixed-effects models
- Longitudinal models
- Missing data analysis
- Multiple sclerosis disease
- Robust models
- Skew-elliptical and elliptical distributions
- Variable selection for mixed models

Grants and contracts

Active

07/2024–06/2027 **NMSS BI-2306-41591** (PI: Schumacher). Epigenetics in MS: An evaluation of biological aging and disease severity. Role: Principal Investigator (30% support).

08/2026–07/2027 **IPR Seed Grant** (PI: Schumacher). More Than a Number: Bayesian Frameworks for Uncertainty Propagation on Epigenetic Clock Predictions. Role: Principal Investigator.

07/2024–06/2029 **NIH/NIA K23 AG084848** (PI: Yinan Zhang). Associations of biological aging with outcomes and immune dysregulation in multiple sclerosis. Role: Co-Investigator (5% support).

Completed

09/2022–05/2025 **NIH/NIA R03AG078946** (PI: Yinan Zhang). Markers of Biological Aging in Multiple Sclerosis. Role: Biostatistician (5–10% support).

08/2018–05/2025 **NIH/NICHD R01HD091274-01A1** (PI: Maria Gallo). Postpartum Family Planning. Role: Co-Investigator (10% support).

Publications

An asterisk (*) marks a student advisee.

Statistical methodology

1. Travassos, K.*, Matos, L. A., **Schumacher, F. L.** (2026). Sandwich variance estimation in scale mixture of skew-normal linear mixed models. *Statistics in Medicine*, 45(23–24), e70753.
2. **Schumacher, F. L.**, Matos, L. A., Ramos, P. L., Louzada, F. (2026). Bayesian estimation in scale mixture of skew-normal linear mixed models using Hamiltonian Monte Carlo. *Brazilian Journal of Probability and Statistics*, in press.
3. **Schumacher, F. L.**, Cutter, G., Burd, C. E., Archer, K., Zhang, Y. (2026). Clocks out of sync: increased epigenetic aging variability in multiple sclerosis. *Journal of the Neurological Sciences*, 485, 125904.
4. Mani, S. K.*, Rempala, G. A., Kenah, E., **Schumacher, F. L.** (2026). Short-term Bayesian influenza forecasting in a hospital environment using a linear Kalman filter. *Journal of Theoretical Biology*, 624, 112422.
5. Oliveira, D. C., **Schumacher, F. L.**, Lachos, V. H. (2026). The use of the EM algorithm for regularization problems in high-dimensional linear mixed-effects models. *Statistical Methods in Medical Research*, 35(2), 240–254.
6. **Schumacher, F. L.**, Matos, L. A., Lachos, V. H. (2025). skewlmm: An R package for fitting skewed and heavy-tailed linear mixed models. *Journal of Statistical Software*, 115(7), 1–32.
7. **Schumacher, F. L.**, Cutter, G. (2025). The ever growing incidence of meta-analyses and reviews. *Multiple Sclerosis and Related Disorders*, 104, 106753.
8. Travassos, K.*, Matos, L. A., **Schumacher, F. L.** (2025). Diagnostic analysis in scale mixture of skew-normal linear mixed models. *Statistica Neerlandica*, 79(1), e70002.
9. Zhong, K.*, **Schumacher, F. L.**, Castro, L. M., Lachos, V. H. (2025). Bayesian analysis of censored linear mixed-effects models for heavy-tailed irregularly observed repeated measures. *Statistics in Medicine*, 44(3–4), e10295.
10. Loor Valeriano, K. A., **Schumacher, F. L.**, Galarza, C. E., Matos, L. A. (2024). Censored autoregressive regression models with Student-t innovations. *Canadian Journal of Statistics*, 52, 804–828.
11. **Schumacher, F. L.**, Matos, L. A., Lachos, V. H., Abanto-Valle, C., Castro, L. M. C. (2024). A censored time series analysis for responses on the unit interval: an application to acid rain modeling. *Sankhya A*, 86, 637–660.
12. **Schumacher, F. L.**, Lachos, V. H., Matos, L. A. (2021). Scale mixture of skew-normal linear mixed models with within-subject serial dependence. *Statistics in Medicine*, 40(7), 1790–1810.
13. **Schumacher, F. L.**, Dey, D. K., Lachos, V. H. (2021). Approximate inferences for

nonlinear mixed effects models with scale mixtures of skew-normal distributions. *Journal of Statistical Theory and Practice*, 15(3), 1–26.

14. **Schumacher, F. L.**, Ferreira, C. S., Prates, M. O., Lachos, A., Lachos, V. H. (2021). A robust nonlinear mixed-effects model for COVID-19 death data. *Statistics and Its Interface*, 14, 49–57.
15. **Schumacher, F. L.**, Lachos, V. H., Vilca-Labra, F. E. (2018). Influence diagnostics for censored regression models with autoregressive errors. *Australian & New Zealand Journal of Statistics*, 60(2), 209–229.
16. **Schumacher, F. L.**, Lachos, V. H., Dey, D. K. (2017). Censored regression models with autoregressive errors: a likelihood-based perspective. *Canadian Journal of Statistics*, 45, 375–392.
17. Souza, E. M., Alves, D. B. M., **Schumacher, F. L.** (2015). Harmonic analysis of multipath index time series in GPS stations. *Tendencias em Matemática Aplicada e Computacional* (*TEMA*), 16, 71.
18. **Schumacher, F. L.**, Souza, E. M., Gimenes, E., Uchimura, T. T., Masukawa, M. L. T. (2014). Modelos estruturais em series temporais com aplicação em dados de internação por bronquiolite. *Revista da Estatística da Universidade Federal de Ouro Preto*, III, 188–199.
19. Philippsen, A. S., Peralta, D., Gazola, S., **Schumacher, F. L.**, Kung, J. (2014). O jornalismo impresso na comunidade universitária. *Extensio* (*Florianópolis*), 11, 16–25.

Statistical applications

1. Zhang, Y., **Schumacher, F. L.**, Clark, L. A., Clark, B. C. (2026). Leg strength and neural activation deficits in older adults with multiple sclerosis. *Multiple Sclerosis and Related Disorders*, 108, 107050.
2. Nassani, L. M., Bencharit, S., **Schumacher, F. L.**, Mandarano, N. A., Place, L. A., Fernandes, G. V. O. (2026). Comparative analysis of dental students' preferences and satisfaction with procedures and outcomes using resin 3D-printing and conventional methods for provisional restoration fabrication. *Dental and Medical Problems*, 63(2), 349–356.
3. Warren, E., **Schumacher, F. L.**, Keder, L., Lawley, M., Goedken, P., Dupper, A. C., Gallo, M. F. (2025). Is 6 weeks postpartum too late for contraceptive counseling in the US? A look into the timing of postpartum intercourse. *Journal of Sex Research*, 1–5.
4. Allen, M. D., Thompson, A., Clark, B. C., **Schumacher, F. L.**, Zhang, Y. (2025). Multiple sclerosis and sarcopenia: analysis of population-level prevalence and key risk factors from the UK Biobank. *Multiple Sclerosis and Related Disorders*, 102, 106629.
5. Vavetsi, K., Mendes, K., Gomes, A. T., Marques, T., Rosa, N., Mandarano, N., **Schumacher, F. L.**, Tatakis, D. N. (2025). TAM pathway receptor proteins: differential expression in healthy human masticatory mucosa. *Journal of Periodontal Research*, 60(11), 1168–1170.
6. Jatana, C., Conley, K., Moffitt, A., **Schumacher, F.**, Emam, H. (2025). Female dental students' perceptions on barriers and motivators in pursuing oral and maxillofacial surgery. *Journal of Dental Education*, 89(8), 1187–1193.
7. Gallo, M. F., **Schumacher, F. L.**, Lawley, M., Keim, S. A., Dupper, A. C., Keder, L. (2025). Potential effect of immediate postpartum use of injectable contraception on lactogenesis. *Contraception*, 142, 110726.
8. Park, G., Meirelles, L., Gouveia, D., Lee, D., Lu, W. E., **Schumacher, F. L.**, Leblebicioglu, B. (2025). Accuracy of dental implant placement with CAD-CAM 3D printed and conventional thermoplastic surgical pilot guides: a clinical comparative trial. *The Journal of Prosthetic Dentistry*, 134(6), 2263–2271.
9. Zaiger, J., Leblebicioglu, B., Meirelles, L., Lu, W.-E., **Schumacher, F. L.**, Tatakis, D. N. (2024). Effects of extraoral storage time on autologous gingival graft early healing: a split-mouth randomized study. *Journal of Periodontal Research*, 59, 1143–1152.
10. Nassani, L. M., Bencharit, S., **Schumacher, F.**, Lu, W. E., Fernandes, G. V. O. (2024). The

impact of technology teaching in the dental predoctoral curriculum on students' perception of digital. *Dentistry Journal*, 12(3), 75.

11. Stanwick, M., Fenesha, F., Hamid, A., Kang, K., Kanniard, D., Kim, I., Mandarano, N., **Schumacher, F. L.**, Peters, S. B. (2024). Impaired tertiary dentin secretion after shallow injury in Tgfbr2-deficient dental pulp cells is rescued by extended CGRP signaling. *International Journal of Molecular Sciences*, 25(13), 6847.
12. Stetzel, L., Foucher, F., Jang, S. J., Wu, T.-H., Fields, H., **Schumacher, F.**, Richmond, S., Ko, C.-C. (2024). Artificial intelligence for predicting the aesthetic component of the index of orthodontic treatment need. *Bioengineering*, 11(9), 861.
13. Wu, Y. C., **Schumacher, F. L.**, Tatakis, D. N. (2024). Participant experience of taking part in periodontal experimental studies. *International Journal of Dentistry*, 2024(1), 8888815.
14. Conley, K. H., Jatana, C., **Schumacher, F. L.**, Lu, W. E., Moffitt, A. (2023). Exploring the attitudes of women dental students regarding becoming oral and maxillofacial surgeons. *Journal of Oral and Maxillofacial Surgery*, 81(9), S30.
15. Donmez, M. B., Çakmak, G., Dede, D. Ö., Küçükekenci, A. S., Lu, W. E., **Schumacher, F. L.**, Revilla-León, M., Yilmaz, B. (2023). Effect of scan powder and scanning technology on measured deviations of complete-arch implant supported frameworks digitized with industrial and intraoral scanners. *Journal of Dentistry*, 138, 104736.
16. Lima, S. G., **Schumacher, F. L.**, Cazarin, C. B. B. (2023). Front-of-package label in Brazil: comparison of guideline daily amounts, traffic-light and warning systems. *Food and Humanity*, 1, 199–209.
17. Yilmaz, B., Dede, D. Ö., Donmez, M. B., Küçükekenci, A. S., Lu, W. E., **Schumacher, F. L.**, Çakmak, G. (2023). Comparison of intraoral and laboratory scanners to an industrial-grade scanner while analyzing the fabrication trueness of polymer and titanium complete-arch implant-supported frameworks. *Journal of Dentistry*, 138, 104697.

Papers submitted or under revision

1. Brunehilde Pinto da Silva, C., **Schumacher, F. L.**, Turkmen, A. A Hamiltonian Monte Carlo approach to Bayesian empirical likelihood in longitudinal data analysis.

Graduate students trained

PhD dissertation advisor

- Runze Cui (expected 2029 Ph.D.). Biostatistics, The Ohio State University.
- Keyliane Travassos Almeida da Silva (expected 2027 Ph.D.). Statistics, University of Campinas. Co-supervision with Dr. Larissa A. Matos.

PhD dissertation committee member

- Vitoria Spinola (2025–). Health Behavior and Health Promotion.
- Amy Dupper (2024–2026). Epidemiology.
- Meredith McCormack-Mager (2025). Biostatistics.
- Yi Tang Chen (2025). Biostatistics.
- Micaela Richter (2025). Biostatistics.
- Danielle Tyson (2023–2024). Epidemiology.

PhD candidacy committee member

- Oksana Kutsa (2026–). Epidemiology.
- Fandi Chang (2025–). Biostatistics.
- Nick Mandarano (2025). Biostatistics.
- Vitoria Spinola (2025). Health Behavior and Health Promotion.
- Amy Dupper (2023–2024). Epidemiology.
- Micaela Richter (2023–2024). Biostatistics.
- Yi Tang Chen (2023). Biostatistics.
- Meredith McCormack-Mager (2022–2024). Biostatistics.
- Danielle Tyson (2022–2023). Epidemiology.

Master's thesis advisor

- Sarah Frost, 2026 MPH, Biostatistics, The Ohio State University. *Social and Geographic Determinants of Rabies Post-Exposure Prophylaxis Administration Following Animal Bites in Allegheny County.*
- Christilene Tumsiah, 2026 M.S., Biostatistics, The Ohio State University. *The Role of Early Diagnosis in Multiple Sclerosis Disease Trajectories: A Multidimensional Analysis of Clinical and Imaging Outcomes.*
- Wenjing Liu, 2026 M.S., Biostatistics, The Ohio State University. *Screening and Selection of DNA Methylation CpGs for Predicting Cognitive Trajectories in ADNI.*
- Dimitra Herbei, 2024 M.S., Biostatistics, The Ohio State University. *Exploring the Association of Multiple Sclerosis and Sarcopenia under Missing Data: A Comparative Exploration Based on the UK Biobank.*
- Keyliane Travassos Almeida da Silva, 2023 M.S., Statistics, University of Campinas. Co-supervision with Dr. Larissa A. Matos. *Diagnostic Analysis in Scale Mixture of Skew-Normal Linear Mixed Models.*

Master's thesis committee member

- Shankar Mani (2023–2024). Biostatistics.
- Jacob Falter (2022–2023). Orthodontics.
- Leah N. Stetzel (2022–2023). Orthodontics.

Master's examination committee member

- Kambra Crist. Biostatistics (MPH, 2026 SU).
- Victoriah Shade. Epidemiology (MPHP, 2026 SP).
- Miriam Donneyong. Epidemiology (MPHP, 2025 SP).
- Akua Amponsah. Epidemiology (MPHPEP, 2024 AU).
- Madison Theodore. Epidemiology (MPHP, 2024 SP).

Undergraduate student supervision

- Syed Rafey Abbas. Ohio Five Summer Undergraduate Research Experience program (2024 SU).
- Yiting Lu. Research Internship (2023 SU+AU), Hangzhou Dianzi University, China.
- Anthony Thompson. The Ohio State University, Undergraduate Capstone Research in Public Health (2023 SP).

Courses taught

The Ohio State University, College of Public Health

Spring 2026	PUBHBIO 7230 — Applied Longitudinal Data Analysis
Spring 2025	PUBHBIO 7230 — Applied Longitudinal Data Analysis
Fall 2024	PUBHBIO 6260 — Ethics in Biostatistics
Spring 2024	PUBHBIO 7230 — Applied Longitudinal Data Analysis
Fall 2023	PUBHBIO 6260 — Ethics in Biostatistics
Spring 2023	PUBHBIO 7230 — Applied Longitudinal Data Analysis

National School of Statistical Sciences (ENCE-IBGE)

Fall 2017	EST065 — Survey methodology (Planejamento de Pesquisas)

Professional presentations

Conferences

1. Flexible joint models for clinical outcomes with skewed and heavy-tailed random effects. **Topic-contributed paper session.** ASA Joint Statistical Meetings 2026, Boston, MA, USA.
2. A robust multilevel framework for modeling longitudinal bilateral manual dexterity in multiple sclerosis. **Oral-contributed presentation.** 33rd International Biometric Conference, Seoul, South Korea.
3. Bayesian robustness for longitudinal data: scale mixture of skew-normal linear mixed models via HMC. **Poster presentation.** 2026 ISBA World Meeting, Nagoya, Japan.
4. Sandwich variance estimation in scale mixture of skew-normal linear mixed models. **Oral-contributed presentation.** ENAR 2026, Indianapolis, IN, USA.
5. Association between epigenetic aging markers and MS disease outcomes. **Poster presentation.** ACTRIMS Forum 2026, San Diego, CA, USA.
6. Censored linear mixed-effects models for heavy-tailed irregularly observed longitudinal data. **Invited session.** 65th ISI World Statistics Congress, 2025, The Hague, Netherlands.
7. Statistical pitfalls in measuring biological aging with epigenetic clocks: insights from a chronic disease setting. **Plenary talk.** 69ª RBras and 21º SEAGRO, Vitoria, ES, Brazil.
8. EMLMLasso: an EM algorithm implementation for regularization in high-dimensional linear mixed models. **Poster presentation.** ENAR 2025, New Orleans, LA, USA.
9. What are we measuring with epigenetic clocks in people with MS? **Poster presentation.** ACTRIMS Forum 2025, West Palm Beach, FL, USA.
10. Assessment of biological aging in people with multiple sclerosis based on epigenetic clocks. **Poster presentation.** 10th Tykeson Fellows Conference, 2024, National Multiple Sclerosis Society, Denver, CO, USA.
11. Challenges in assessing biological aging in people with multiple sclerosis based on epigenetic clocks. **Invited session.** 37th New England Statistics Symposium, 2024, University of Connecticut, Storrs, CT, USA.
12. Bayesian scale mixture of normal censored linear mixed models with within-subject serial dependence. **Invited session.** CMStatistics 2023, Berlin, Germany.
13. Variable selection in high-dimensional linear mixed-effects models. **Contribución científica.** Vigésimas Jornadas en Estadística, 2023, Guayaquil, Ecuador.
14. Panel on how to join and explore pharma. **Panelist.** Statistics in Pharmaceuticals, 2023, Storrs, CT, USA.
15. Robust mixed models for longitudinal data: model fitting and evaluation. **Speed presentation.** ASA Joint Statistical Meetings 2023, Toronto, Canada.
16. EM algorithm for regularization problems in high-dimensional LMMs. **Flash talk.** IMS 23rd Meeting of New Researchers in Statistics and Probability, 2023, Toronto, Canada.
17. Bayesian estimation in scale mixture of skew-normal linear mixed models using Hamiltonian Monte Carlo. **Oral presentation.** ISI 64th World Statistics Congress, 2023, Ottawa, Canada.
18. Censored autoregressive regression modeling using the R package ARCensReg. **Oral presentation.** 24th International Conference on Computational Statistics, 2022, Bologna, Italy.
19. Modeling longitudinal data using robust mixed models in R. **Tutorial.** 24º Simpósio Nacional de Probabilidade e Estatística (SINAPE), 2022, Gramado, RS, Brazil.
20. Censored autoregressive regression models with Student-t innovations. **Oral presentation.** ICSA 2022 Applied Statistics Symposium, 2022, Gainesville, FL, USA.
21. Canonical fundamental skew-t linear mixed models. **Oral presentation.** III Jornadas Ecuatorianas en Matemáticas, 2021, Ecuador (virtual).
22. Scale mixture of skew-normal linear mixed models with within-subject serial dependence. **Poster presentation.** 34th New England Statistics Symposium, 2021, Providence, RI, USA.

23. skewlmm: an R package for robust estimation of linear mixed models based on scale mixture of skew-normal distributions. **Poster presentation.** ISI World Statistics Congress 2021.
24. Canonical fundamental skew-t linear mixed models. **Oral presentation** (Student Prize Session). Conference in Honour of Fred Smith and Chris Skinner, 2021 (virtual satellite meeting of the ISI World Statistics Congress).
25. Scale mixture of skew-normal linear mixed models with within-subject serial dependence. **Oral presentation** (Organized Invited Session). 4th International Conference on Econometrics and Statistics, 2021, Hong Kong (virtual).
26. Robust estimation in linear mixed models using the R package skewlmm. **Oral presentation.** XV Encontro Científico de Pós-Graduandos do IMECC, November 2020, Campinas, SP, Brazil (virtual).
27. Autoregressive scale mixture of skew-normal linear mixed models. **Poster presentation.** VI Workshop em Análise de Sobrevivência e Aplicações (WASA), November 2019, Piracicaba, SP, Brazil.
28. Influence diagnostics for censored regression models with autoregressive errors. **Oral presentation.** XVI Escola de Modelos de Regressão (EMR), March 2019, Pirenópolis, GO, Brazil.
29. Censored regression models with autoregressive errors. **Oral presentation.** 23º Simpósio Nacional de Probabilidade e Estatística (SINAPE), September 2018, São Pedro, SP, Brazil.
30. *Imputação de dados faltantes em séries temporais.* **Poster presentation.** XIV EMR, March 2015, Campinas, SP, Brazil.
31. *Modelos estruturais em séries temporais com aplicação em dados de internação por bronquiolite.* **Oral presentation.** 59ª RBras, May 2014, Ouro Preto, MG, Brazil.

Guest lectures

1. Guest speaker discussing difference-in-differences models. VETPREV 8725: Quantitative Methods Capstone, The Ohio State University, February 20, 2025.

Other invited presentations

1. Beyond normality: flexible linear mixed models for repeated measurements. **Seminar.** Quantitative Psychology Program, Department of Psychology, The Ohio State University, Columbus, OH, USA, September 21, 2026.
2. Flexible longitudinal models and epigenetic aging markers in multiple sclerosis. **Seminar.** Department of Biostatistics & Informatics, Colorado School of Public Health, University of Colorado Anschutz Medical Campus, Aurora, CO, USA, September 9, 2026.
3. The use of AI in research and teaching of biostatistics at OSU. **Seminar.** Department of Mathematics and Statistics, Federal University of São João del-Rei, São João del-Rei, MG, Brazil, December 11, 2025.
4. Challenges in longitudinal data analysis: complex made simple with the R package skewlmm. **Seminar.** 2024 Fall Speaker and Annual Banquet, Columbus ASA Chapter, Columbus, OH, USA, September 16, 2024.
5. Lessons from an ethics in biostatistics course. **Seminar.** Department of Statistics, University of Campinas, Campinas, SP, Brazil, July 12, 2024.
6. Scale mixture of skew-normal linear mixed models: a Bayesian approach using Hamiltonian Monte Carlo. **Seminar.** Department of Bioinformatics and Biostatistics, University of Louisville, Louisville, KY, USA, January 26, 2024.
7. Penalized estimation of scale mixture of skew-normal linear mixed models using Hamiltonian Monte Carlo. **Seminar.** Division of Biostatistics and Bioinformatics, University of Cincinnati, October 27, 2023.
8. Penalized estimation of scale mixture of skew-normal linear mixed models using Hamiltonian Monte Carlo. **Seminar.** Department of Statistics, The Ohio State University, October 19, 2023.

9. Penalized estimation of scale mixture of skew-normal linear mixed models using Hamiltonian Monte Carlo. **Colloquium.** Department of Statistics, Federal University of Minas Gerais, September 22, 2023.
10. A trajetória acadêmica e uma introdução a modelos mistos para dados longitudinais. **Invited lecture.** Department of Statistics, State University of Maringa, March 23, 2022.
11. Robust mixed-effects models for longitudinal data. **Colloquium.** Department of Statistics, University of Connecticut, November 17, 2021.

Software

1. **Schumacher, F. L.**, Matos, L. A., Lachos, V. H. (2020). skewlmm: Scale mixture of skew-normal linear mixed models. https://cran.r-project.org/package=skewlmm
2. Zhong, K., **Schumacher, F. L.**, Castro, L. M., Lachos, V. H., Carrasco, J. M. F. (2025). SMNlmec: Scale mixture of normal distribution in linear mixed-effects model. https://cran.r-project.org/package=SMNlmec
3. **Schumacher, F. L.**, Lachos, V. H., Galarza, C. E. (2016). ARCensReg: Fitting univariate censored linear regression model with autoregressive errors. https://cran.r-project.org/package=ARCensReg

Professional service

Conference and workshop organization

- Statistics in Pharmaceuticals 2023. University of Connecticut, Storrs, CT, USA. Member of the Organizing Committee.
- XIII Encontro Científico dos Pós-Graduandos do IMECC, 2018. Campinas, SP, Brazil. Member of the Organizing Committee.

Associate editor

- *Sankhya B*, 2023–2026.

Professional societies

- American Statistical Association, Columbus Chapter — Communication Officer, 2025–present.
- Caucus for Women in Statistics — Award Committee member, 2025–present.

Service to the profession

- MS Australia, Incubator Grant Reviewer, 2026.
- Patient-Centered Outcomes Research Institute (PCORI), Methods Merit Reviewer, 2025.
- OSU Florence Nightingale Day for Statistics and Data Science, activity leader, October 2023, October 2024, and October 2025.
- OSU TDAI Data Science Summer Camp, panelist, June 2023.

Referee service

- Biometrics
- Brazilian Journal of Probability and Statistics
- Communications for Statistical Applications and Methods
- Communications in Statistics — Case Studies and Data Analysis
- Environmental and Ecological Statistics
- Frontiers in Neurology
- IEEE Transactions on Industrial Informatics
- Infectious Disease Modelling
- JAMA Neurology
- Journal of Alzheimer's Disease
- Journal of Applied Statistics
- Journal of Multivariate Analysis
- Journal of Scientific Research and Reports
- Journal of Statistical Computation and Simulation
- Journal of Statistical Theory and Practice

- Journal of the Korean Statistical Society
- Journal of the Royal Statistical Society: Series C
- Neurology
- R Journal
- Sankhya B
- Statistics
- Statistics and Computing
- Statistics in Medicine

Professional affiliations

- American Statistical Association
- Associação Brasileira de Estatística (ABE)
- Caucus for Women in Statistics
- International Statistical Institute
- Região Brasileira da Sociedade Internacional de Biometria (RBras)

Online profiles

ORCID	https://orcid.org/0000-0002-5724-8918
Google Scholar	https://scholar.google.com.br/citations?user=WNtJVuUAAAAJ
OSU profile	https://cph.osu.edu/people/fschumacher
Personal page	https://sites.google.com/view/fernandalschumacher/